Variance and Standard Deviation

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Introduction

Variance and standard deviation are the most widely used dispersion measures in applied statistics. The variance is the mean squared deviation from the mean; the SD is its square root, returning to the original units. Both underlie the normal distribution, the central limit theorem, regression, and every parametric test that refers to "sigma".

Prerequisites

The reader should know the mean and be comfortable squaring, summing, and taking square roots in R.

Theory

For a sample $x_1, \ldots, x_n$:

- **Population variance** (when the data *are* the whole population): $\sigma^2 = \frac{1}{n} \sum (x_i - \mu)^2$.
- **Sample variance** (when the data are a sample): $s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$.

The $n - 1$ divisor (Bessel's correction) makes s^2 an unbiased estimator of σ^2. Using n would systematically underestimate σ^2 because $\bar{x}$ is itself estimated from the same data.

The sample **standard deviation** is $s = \sqrt{s^2}$. Even though s^2 is unbiased, s is biased downward because the square root is concave (Jensen's inequality). The bias shrinks like $1/n$ and is negligible for $n > 30$; an unbiased estimator exists (`c_4` correction in SPC) but is rarely used in applied work.

R's `var()` and `sd()` use the $n - 1$ divisor by default. To get the MLE version, compute `mean((x - mean(x))^2)`.

Assumptions

Variance and SD are descriptive; no distributional assumption is required. For inferential use (the standard error of the mean is $s/\sqrt{n}$), the usual iid assumption applies.

R Implementation

```r
set.seed(2026)
x <- rnorm(60, mean = 100, sd = 15)

var(x)
sd(x)

# MLE (biased) version with divisor n
mean((x - mean(x))^2)
sqrt(mean((x - mean(x))^2))

# Bias of s: theoretical factor c_4(n) for normal data
c4 <- function(n) sqrt(2 / (n - 1)) * gamma(n / 2) / gamma((n - 1) / 2)
```

```r
sd_unbiased <- sd(x) / c4(length(x))
sd_unbiased
```

Output & Results

```
[1] 210.5       # s^2 with divisor n-1
[1]  14.5       # s
[1] 207.0       # MLE variance
[1]  14.4       # MLE SD (biased)
[1]  14.6       # bias-corrected SD
```

At $n = 60$, the difference between biased and unbiased SD is a fraction of a percent, irrelevant for applied work.

Interpretation

Report variance when the scale of the squared units is interpretable (e.g., variance of portfolio returns). Report SD when the scale of the original units is desired. In a paper: "mean 100.2 (SD 14.5)" is universally readable.

Practical Tips

- Always specify which divisor you mean when reporting; R's default is $n - 1$, which is the statistician's convention.
- For very small samples, the biased-vs-unbiased distinction matters; for most applied work (n > 30) it does not.
- The SD is in the same units as the data; the variance is in units squared. Don't mix them in a single report.
- When a variance is reported without an associated mean, check whether it is actually a standard error (which is smaller by $\sqrt{n}$).
- For skewed data, the SD is not a useful summary; use the IQR instead.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Descriptive statistics and Table 1